DiracX developments

Directions

Alexandre F. Boyer · CERN

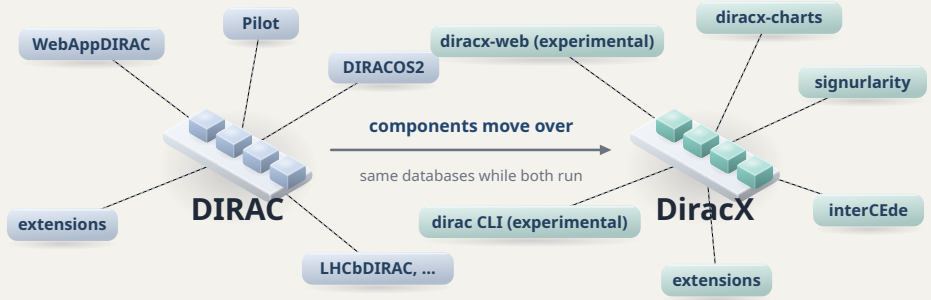
12th DIRAC(X) Users' Workshop · FZU Prague · 13–16 October 2026



1 · Where we are now

DIRAC and DiracX today

Maintaining DIRAC, progressing on DiracX



DIRAC v9.1.21 · v9.0.27 · v8.0.85
WebAppDIRAC 6.0.6 · DIRACOS2 2.63
Patched, not developed: security fixes, and performance fixes when needed.

DiracX v0.6.0: foundations, authentication, jobs (search, status), sandboxes, read-only resource status, and experimental observability (OpenTelemetry).
The web app (diracx-web 0.1.0-a11) and the `dirac` CLI are still experimental.

> **Adopt what already exists in DiracX.** DIRAC is only patched, so that DiracX can progress. LHCb runs one service and one task in production; two other installations are certifying DiracX.

2 · Evolving requirements, fruitful collaboration

New communities, new use cases, and two systems running side by side

What DIRAC is for, and what it values

> Running large-scale workflows of embarrassingly parallel jobs on heterogeneous, distributed computing and storage resources: high-throughput computing.

Minimal dependencies

A database and a search index, on a few hosts.

In DiracX: standard protocols for what is external (S3, OpenTelemetry, OIDC). Without Kubernetes: Thursday's talk.

Multi-VO installations

One installation serves several communities.

In DiracX: multi-VO from the start.

Extendable

Communities add their own pieces.

In DiracX: extensions, with a reference extension.

My own reading of what communities have asked for over the last few years.

Requirements keep evolving

New communities

CMS and FCC, who joined this summer, bring new requirements. CMS has no DIRAC installation and runs its own workload management: Wednesday's CMS talk.

Advanced needs, in extensions

CTAO, LHCb, Belle II and JUNO built extensions for advanced needs that are now close to each other. They belong in the common project: CTAO's use of **CWL** turned out useful for DiracX as a whole.

Constant needs

GridPP has few resources: DiracX must stay very simple to deploy, with minimal dependencies.

> Share your needs early, at DOps, before building around the system: someone else may need the same thing.

What stays stable, what is new

Stable

- **DIRAC keeps running**, with security and performance patches.
- Components move **one at a time**, over the same databases.
- **Extensions keep working** while components move.

New

- **The transformation system**: a new data model, new logic, and a much richer scope than DIRAC's.
- **Decoupled from DIRAC**, so DIRAC's databases need no large changes and stay stable.
- **Standard protocols** (sandboxes in S3, observability in OpenTelemetry) and multi-VO from the start.

> A stable path for what exists, and room for what is new.

Many part-time hands



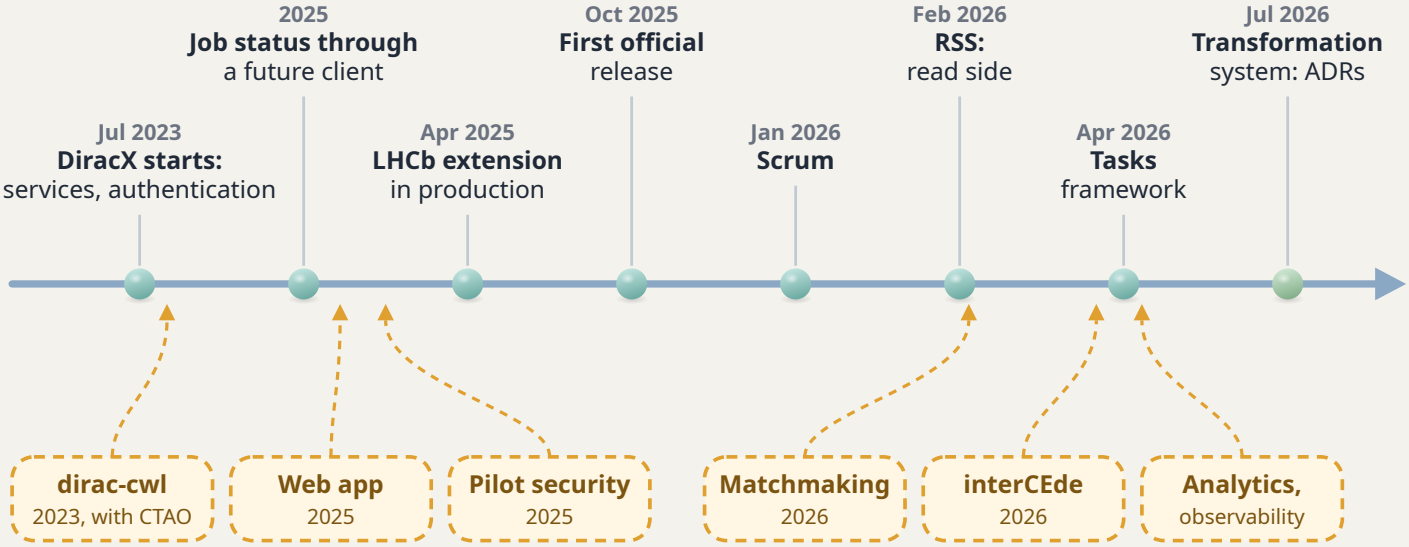
- **36 people** contributed since the last workshop, **10 to 18** in any given month, almost all part-time: about **5 to 6 full-time equivalents** in total, LHCb core developers included.
- They maintain DIRAC and develop DiracX at the same time.

> **Finding the best way to work with each other is paramount.**

3 · Adapting quickly

Responding to changing requirements, in small steps

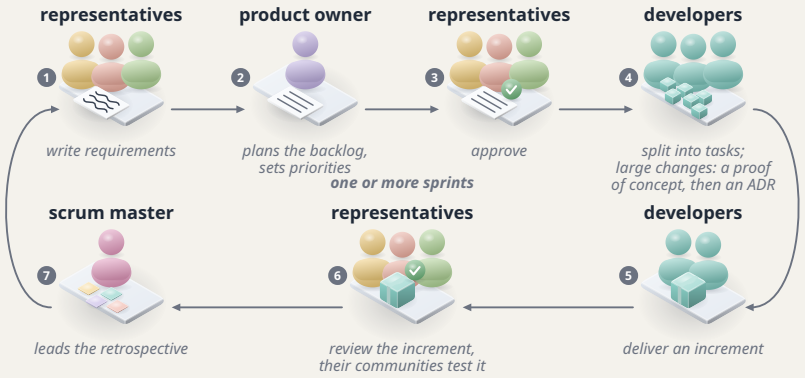
The journey so far



In parallel: proofs of concept, linked to when they started

- **From 2023, three hackathons a year** onboarded new contributors, with no dashboard or regular meeting to follow progress. **Scrum** brought both in 2026.
- **Proofs of concept run in parallel**, not mandatory: we try an idea before committing to it.

Almost the current process



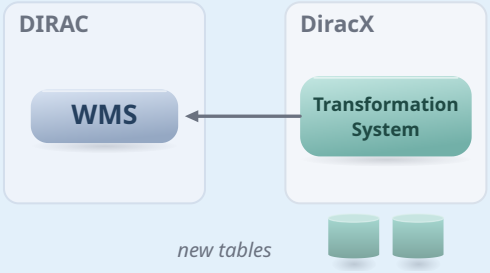
- **Scrum since 21 January 2026**: two-week sprints; planning and refinement in one weekly meeting; a monthly meeting with the representatives (DOps); hackathons.
- **Large changes start with a proof of concept, then an ADR** that the other developers approve. ADRs and roadmap are public; the board is not.

> **Individuals and interactions over processes and tools.** The process is a guide, not a rule, and it adapts as we learn.

Not a port: a reimplementation

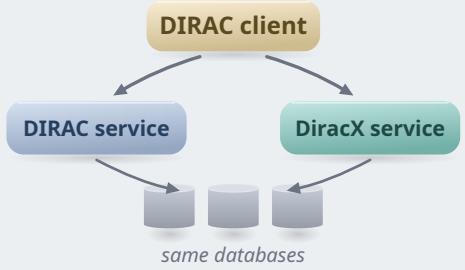
A new data model

The spine: workgraph, transformations, parcels. New requirements need new tables and new logic. The RMS is used as it is, as a backend.



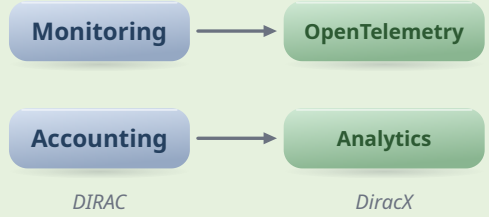
Future clients

DMS: a DIRAC client calls DiracX instead of DIRAC, over the same databases. **RSS**: the read side is in DiracX; DIRAC's adapter is in review.



Standard protocols

Systems internal to DIRAC give way to standards: monitoring to OpenTelemetry, accounting to analytics.



This week: Data Management for DiracX · Analytics for DiracX · RSS in DIRAC and DiracX (all Wednesday)

Delivered since the last workshop

DiracX, in use

- From alpha to **22 releases**: the first official one on 22 October 2025, **v0.6.0** on 8 October 2026.
- **LHCb in production**; two other installations certifying.

Usable now

Sandboxes in S3, **job status** through the future client, **resource status** read in DiracX, the **tasks framework**, and experimental **OpenTelemetry** observability.

DIRAC kept safe

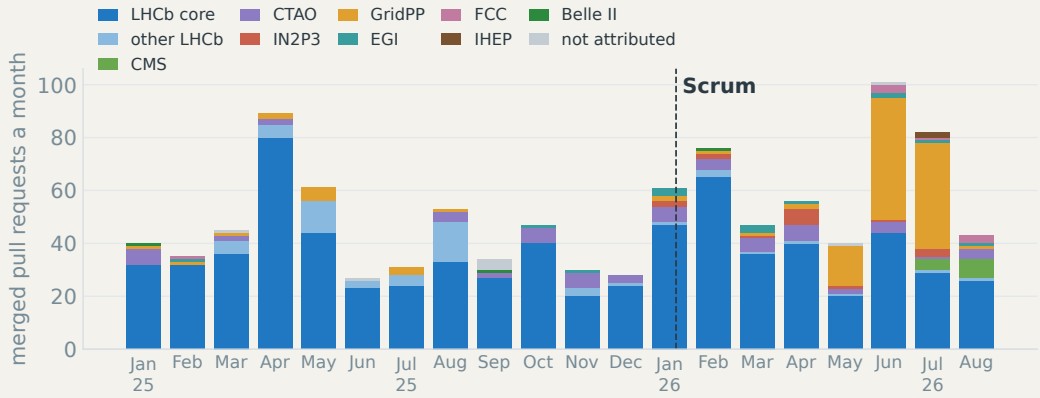
v8, v9.0 and v9.1 patched: **security fixes** and **SQL hardening**, with GridPP.

Your needs, handled. CMS and FCC joined, and their requirements reordered the roadmap. Requirements from five communities became **eight transformation-system ADRs**, reviewed together. **CWL** went from a proof of concept with CTAO to the project's direction. **268 issues** closed as completed.

4 · How we work, measured

What the numbers tell us about our process

More hands, from more communities



634 pull requests merged across all DIRACGrid repositories since the last workshop (367 DIRAC, 261 DiracX), from nine communities; **17 new contributors**. Almost all are part-time, so effort varies from month to month.

Outside the five core developers: **36% of merged PRs, 28% of issues opened, 53% of story points.**

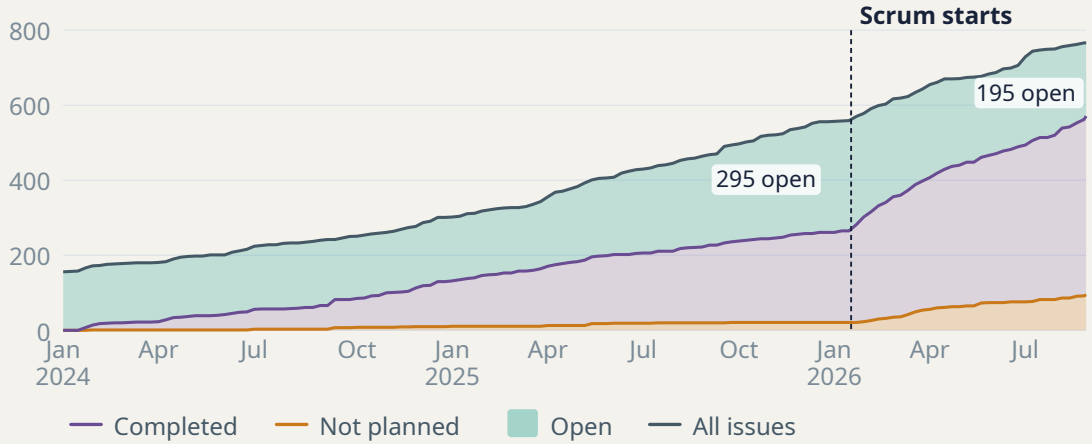
Read with care: about half of the 2026 rise in PRs is **one GridPP developer** hardening DIRAC's SQL. CMS merged 14 changes since July.

Since Scrum: the backlog is shrinking

295 → 195
open issues, January 2026 to September

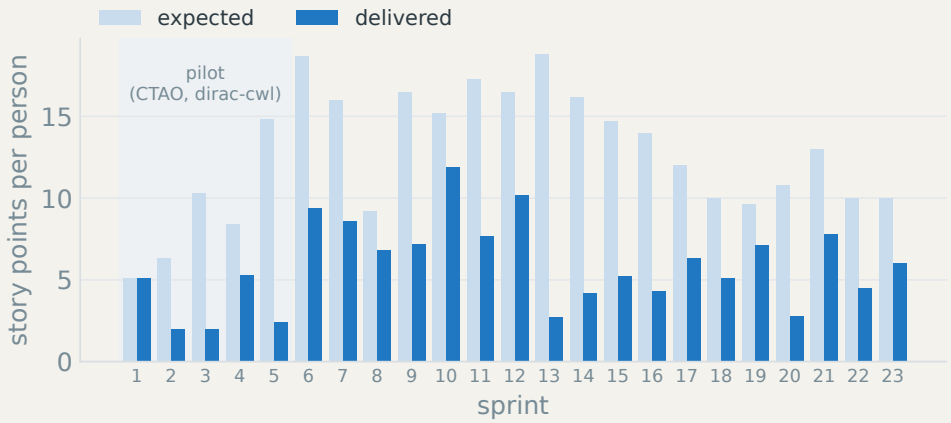
10.6 → 36.6
closures a month; 26.6 completed, 10.0 not planned or duplicate

16.3 → 27.3
issues opened a month



All issues in the DIRACGrid repositories, bots excluded. Part of the fall is triage, and about 120 of the 195 still open are not triaged yet.

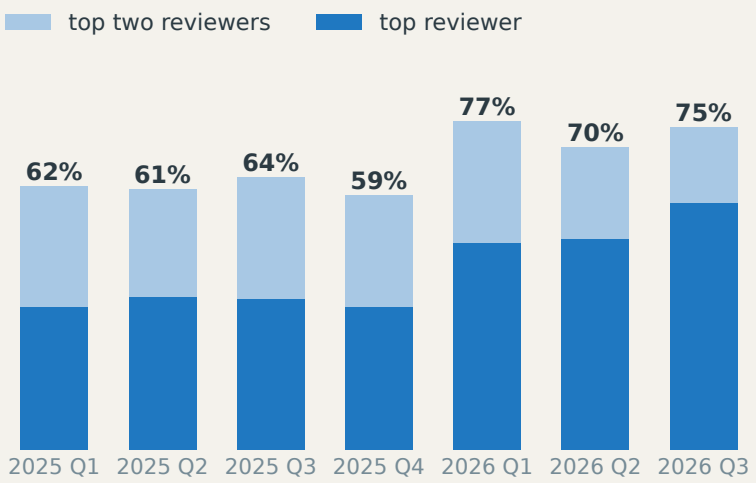
We plan for twice what we deliver



- Since January, a median of **14 story points** per person and sprint expected, **6.6 delivered**. No sprint delivered what was expected.
- Delivered velocity follows availability: workshops, conferences, holidays.
- **Unplanned work lives in DIRAC**: 56% of the story points delivered on DIRAC were unplanned, against 11% on DiracX.

For developers: we overestimate what a sprint can hold, partly because we do not count the unplanned DIRAC maintenance. Plan for it.

The bottleneck is review



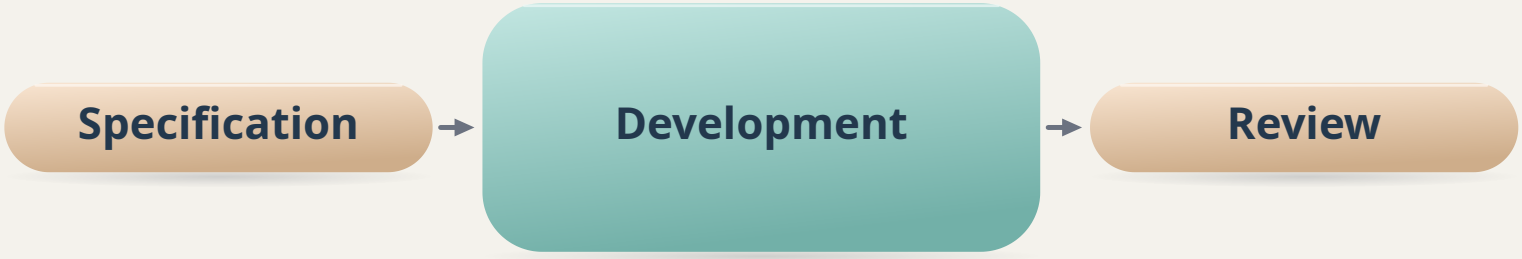
Share of first reviews, by quarter

- **Two people give three quarters of first reviews.** None of the 14 contributors new since January has given one yet.
- **Smaller pull requests get reviewed sooner**: 2.6 hours under 10 lines, 23 hours from 500 lines. On small changes the core five wait 2.5 hours, everyone else 11.
- Design review (ADRs, in meetings and on chat) is not in these numbers.

Question for developers: kanban style, should you review before starting something new once your tasks are done?

How to split a large change into stacked pull requests

Where work waits



- **Scrum widened development**: more contributors, a shrinking backlog.
- **Specification and review are the narrow parts**, and they will matter more with every new component.

For everyone in the room: representatives, developers and users, where would you improve how we work?

5 · What’s next

How we could improve, and the roadmap

Cheaper code: easier to explore, same bottlenecks

> *"AI's primary role in software development is that of an amplifier."*

DORA 2025 State of AI-assisted Software Development

- Writing code got cheap, so **spikes and proofs of concept** are easier to try: matchmaking, interCEde, analytics, observability.
- **Deciding what to build and checking it** did not get cheaper.
- It does **not change our process**. It shows its weak points sooner.

Each one is something we just saw

- **Vague specification** → verbose, AI-generated issues that go stale (an open action). A clear ADR with a scenario test is what an agent, or a newcomer, can implement.
- **Review bottleneck** → more code means more to read, by the same two people.
- **Unplanned work** → faster code does not shrink urgent maintenance.
- **A new kind of user** → DiracX speaks HTTP and OAuth2, so an agent can use it through MCP.

→ *invest in* **specifications, tests and review**, *not in typing*

Next: building the transformation system

Implementation starts at the hackathon this afternoon. The technical details are in the next talk.

Where it comes from

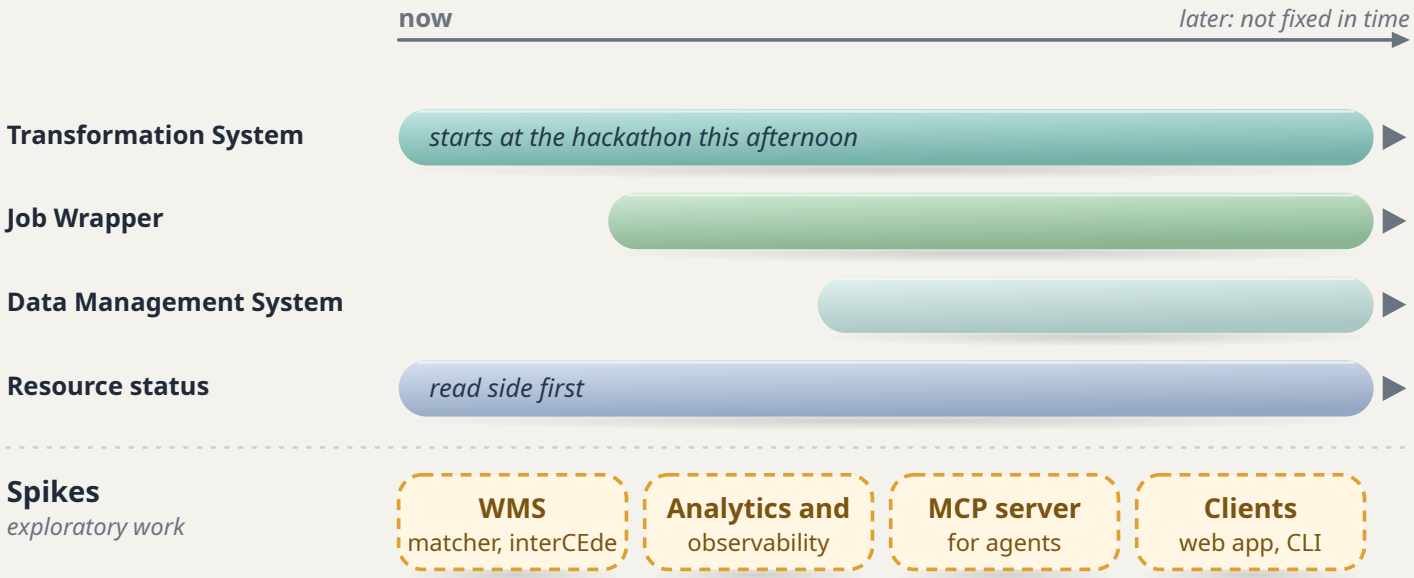
- **Use cases from the communities**: CMS, CTAO, FCC, Belle II and LHCb.
- **A design that comes from a proof of concept**, written down as ADRs.

How it will be delivered

- **Increments the representatives can test**, shown at DOps whenever a sprint delivers one.
- **The more we are, and the better organised, the sooner.**

This afternoon: adapting your community's workflows to **CWL** from 14:00, then building the **transformation system** on them from 16:10.

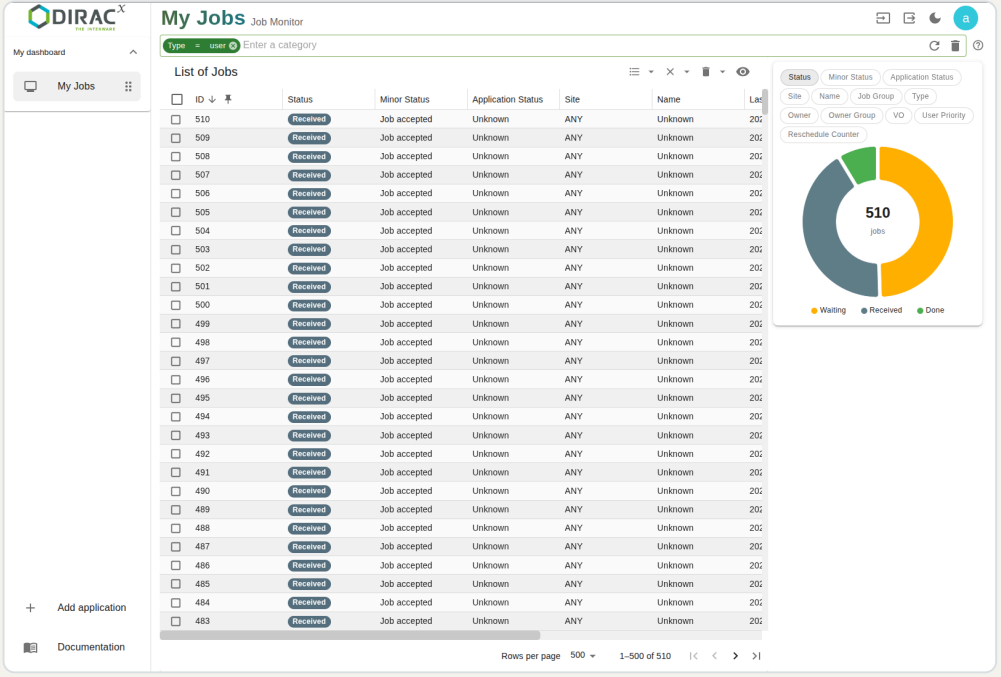
The roadmap moves with the communities



Priorities, not commitments. New requirements can reorder this page, and the team writes the detailed plan.

A word on the web app

Announced at past workshops, not on this week's agenda.



- **Today: a job monitor**, smooth and responsive; extensible, unit and end-to-end tested.
- **We promised a simple app on popular libraries.** AI changes the game: richer interfaces from simple components, fewer dependencies.
- **Next: a consistent, user-centric experience** around the new transformation system and its components.

> **Increments at DOps** once the transformation system is usable. Feedback welcome any time.

If you run an installation today

What you can count on

- **DIRAC keeps running**, with security patches and performance fixes when needed.
- **You choose when to switch each component**, usually at a shutdown.
- **No data migration** for anything but the transformation system, which has its own tables.
- **Your extension keeps working** on DIRAC while you move.
- **DIRAC and DiracX live in parallel**, and work can move gradually from one to the other.
- **Increments to test**, shown at DOps whenever a sprint delivers one.

Take-aways

FOR INSTALLATIONS

- **Adopt what exists; DIRAC is only patched.** You move one component at a time, and the transformation system runs in parallel.

FOR REPRESENTATIVES AND DEVELOPERS

- **Specification and review are the bottlenecks.** Increments you can test, shown at DOps, keep us close to your needs.

FOR THIS AFTERNOON

- **Join the hackathon**: your workflows in CWL from 14:00, then the transformation system from 16:10.

Thank you

Questions and comments welcome, this week and after.

Alexandre F. Boyer · alexandre.boyer@cern.ch

github.com/DIRACGrid · diracx.diracgrid.org

